

Shivam Murawala

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EDUCATION

Woburn Collegiate Institute

Ontario Secondary School Diploma

Toronto, ON

Sep. 2023 – Jun. 2027

- Grade 11 Average: **97.5%**
- **Activities:** Computer Science Club, Math Team, DECA, Robotics, Cross Country Team

EXPERIENCE

Mobile Infrastructure Developer

Nokia

Ottawa, ON

Jul. 2026 – Present

- Resolved 5+ Java bugs within the Nokia Policy Controller (NPC) codebase in under three weeks.
- Onboarded a team of 10+ engineers onto a centralized machine learning platform, and configured 2 VMs to reduce manual setup time by 20%.
- Updated legacy documentation guides for the local developer environment, streamlining onboarding for future Mobile Infrastructure Hires.

Founder and President

WCI Computer Science Club

Toronto, ON

Sep. 2025 – Present

- Founded and led a 40+ member club, driving active participation and successful completion of group projects through weekly web development and programming sessions.
- Led the planning and execution of a school-wide, 2-day hackathon, fostering student collaboration in technology.

Independent Researcher

Toronto Science Fair

Toronto, ON

Sep. 2023 - Mar. 2024

- Conducted 400+ automated simulations comparing leading chess engines, including Stockfish and Leela Chess Zero
- Analyzed differences between Minimax-based search and MCTS/neural-network architectures in gameplay
- Concluded that neural-network engines prioritize long-term positional advantages, while Minimax-based engines excel at tactical refutations

PROJECTS

Conway's Game of Life Simulator 🧬 | C++, Raylib

- Built a high-performance implementation of John Conway's Game of Life using C++ and Raylib
- Implemented real-time controls to start, stop, and modify the simulation state dynamically
- Added support for random grid generation, custom initial states, adjustable simulation speed, and cell colors
- Optimized rendering performance to support large grid sizes smoothly

AWARDS & ACHIEVEMENTS

- **Lloyd Auckland Math Workshop @ University of Waterloo** — One of 95 students invited across Canada
- **Canadian Computing Competition** — Top 10% and School Champion
- **AIME Qualifier: Top 10% on AMC** — Scored 114/150 on AMC 12B
- **DECA International Career Development Conference** — Placed 3rd globally out of 1,000+ competitors
- **FIRST Robotics Competition World Championship** — Programming (Java) and Impact Team, Team 188
- **CEMC Galois Contest** — Distinction, Group V Placement, School Champion
- **CEMC CIMC** — Distinction, Group V Placement, School Champion
- **Math Kangaroo Contest** — National Silver (13th in Canada, 4th in GTA)
- **Toronto Science Fair** — Silver Medal Recipient

TECHNICAL SKILLS

Languages: C++, Python, Java, JavaScript/TypeScript, HTML5, CSS, SQL

Frameworks & Libraries: WPILib, React.js, Matplotlib, Pandas, Scikit-Learn, Raylib, JUnit5

Tools: Git, GitHub, VS Code, Gradle, Figma, Gerrit, Jira, Linux Terminal